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# SDC Hackathon

## Phase 1 : Preliminary Project Proposal

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Revised for the Final Round: 31 October 2026**

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

***Instagram:*** @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

**SDC HACKATHON 2026**

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Hosted by Xiamen University Malaysia (XMUM) AI Club
Revised 31 October 2026 (Final Round deadline: 1 November 2026)

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## Revision summary: what changed since 25 October 2026 and why

The Domain 1 Finalist Guide set the final standard: all six must-haves shown live, trend data with source and date that changes the results, explainable scores, real low-cost courses, a reason on every SDG tag, and scenarios driven by parameters. Every change below answers one of those points, or a gap we found in our own tests.

| What changed | Phase 1 (25 Oct) → Final (31 Oct) | Why |
|---|---|---|
| Order of the product | Before: Where can I move?<br>Now: **Stay first, then move**: which of my skills AI takes, how to stay AI-ready, then where to go | A worker's first question is "will AI take my job, and which part of it?" |
| Skill view | Before: Matched skills and gaps<br>Now: Every skill grouped as at risk, AI-assisted or still yours, with one action each | Gives a next step per skill, not only a score |
| Jobs with routes | Before: 16 sample jobs<br>Now: Any of 3,039 ESCO occupations, with AI-era skills, reviewed cross-field bridges and practice tasks | The sample catalogue could not answer "check my job" |
| Learning plan | Before: 30 listed courses<br>Now: Plus 45 hand-checked learning resources, 38 of them free; every link opened on 30 | The guide requires real free or low-cost resources |

| | Oct | |
|---|---|---|
| Map | Before: Distance = similarity<br>Now: Distance = months to job-ready (3/6/12/24-month rings) | Time matters more to a worker than similarity |
| Scenarios | Before: Steady / Fast toggle<br>Now: Today vs 2030 Slow / Steady / Fast side by side, from one AI-speed parameter, with what shifts spelled out | The guide asks for parameter-driven scenarios and a visible shift |
| Evidence review | Before: One AI reviewer plus fixed rules<br>Now: Three voter agents on models from two vendors, with an audit log; a person decides split votes | A single reviewer accepted vendor marketing; a sceptic voter catches it |
| SDG mapping | Before: Tags on 29 skills and 16 jobs<br>Now: Every ESCO job and skill tagged with a goal and a reason; the path's contribution shown in Plan | The guide requires a reason per tag and a path total |
| Progress | Before: Saved plan<br>Now: My goal: weekly check-ins, proof of each skill, opt-in HR team progress | Learner motivation and the HR view are good-to-haves |
| Thin profiles | Before: CV or pasted text<br>Now: Plus a no-CV task quiz | Robustness for a thin profile |
| AI models | Before: Gemini<br>Now: KrackedDevs proxy (GPT-5.5, gpt-6-sol, DeepSeek-V4-Pro); Gemini when that key is not set | Free-tier quotas; a vote panel on two vendors' models |
| Tests | Before: 418<br>Now: 543 (36 files) | New features are covered |

Final-round results against the Phase 1 targets are reported in Section 4 of the Final Project & Prototype Report (SDC2026_Dsaise_Final.pdf).

## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | Dsaise |
| **Selected Domain** | Domain 1: Aiversecity Future-Ready Workforce |
| **Selected SDG Track (1–17)** | SDG 8 (Decent Work and Economic Growth), primary; SDG 4.4 (skills for employment) |
| **Project / Solution Name** | HalaTuju |
| **Name (Leader)** | Leow Shen En |
| **Email & Phone (Leader)** | leowshenen@gmail.com, 018-2335269 |
| **University Affiliation(s)** | Universiti Malaya |
| **Number of Team Members** | 5 |

## 2. Executive Summary & Problem Statement

### 2.1 The Problem Context

*Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.*

***[Maximum 250 words]***

Mid-career workers need to know which of their skills AI is taking over, how to stay useful in their current job, and where to move if they must. A sample worker in a Penang shared-services centre has six hours a week to learn and a six-month redeployment window. She needs to know which skills still transfer, which new role is realistic, and whether the advice will stay sound as AI changes. Her HR team must also compare openings, training time and cost. One-off career assessments and job-title matching miss transferable skills and changing task exposure, and training lists go out of date before they are finished. This matters at scale: DOSM recorded 1,360,100 Malaysian clerical wage recipients in 2025, a broad group that includes similar workers (OpenDOSM, Annual Salaries and Wages by Occupation and Sex, 2025). The ILO refined GenAI exposure index reports especially high exposure for clerical work (Working Paper 140, 2025). These figures describe context, not a measured displacement rate. HalaTuju supports upskilling and redeployment under SDG targets 8.5 and 4.4.

### 2.2 The Proposed AI Solution

*Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.*

***[Maximum 300 words]***

HalaTuju turns a CV, pasted text or a short task quiz into a skills profile. Local OCR reads scans, and a privacy scrubber removes personal details before any AI model sees the text. Language models (GPT-5.5 and DeepSeek-V4-Pro through the KrackedDevs proxy, with Google Gemini as fallback) and bge-m3 embeddings on RunPod match the CV's evidence against all 13,939 ESCO skills. On 36 held-out synthetic CVs, the current job was identified correctly in 97.2% of cases; the scrubber removed 99.4% of planted personal details after fixes.

Our own model, trained on a RunPod GPU from human task ratings, scores each skill's AI exposure. The worker sees skills at risk, skills AI assists and skills that stay theirs, with one action each. They can stay and become AI-ready, or move: any of 3,039 ESCO occupations gets a ranked match score with reasons, a skills gap, and an ordered route fitted to weekly hours, with free or low-cost resources whose links were checked by hand.

Fixed formulas, not the language model, set every score. Each plan is shown under Today and 2030 Slow, Steady and Fast AI adoption, and replayed in 1,000 simulated futures, labelled as estimates. Every job and skill carries an SDG goal with a reason, and the plan shows its contribution.

A scheduled researcher reads 74 trusted sites and checks every quotation. Three AI agents on two vendors' models vote on each claim; a person decides split votes. Approved changes reroute affected users, who choose to keep or switch, and every change can be undone. Workers track progress weekly; HR teams see opt-in team progress. The app is built with Node and Express, uses PostgreSQL, and is deployed on Replit.

## 3. Technical Architecture & Implementation Plan

### 3.1 Conceptual Tech Stack

*List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.*

*Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.*

Browser UI: JavaScript, HTML/CSS, a 3D career map, English and Bahasa Melayu. API: Node 22, TypeScript, Express 5; local Tesseract OCR and a deterministic scoring engine shared by server and browser. AI: KrackedDevs LLM proxy (GPT-5.5, gpt-6-sol, DeepSeek-V4-Pro), with Google Gemini as fallback; BAAI/bge-m3 embeddings on RunPod Serverless; a team-trained bge-m3 plus ridge exposure model trained on a RunPod GPU pod. Data: PostgreSQL versioned knowledge base; ESCO, O*NET, ILO, DOSM, MyCOL, hand-checked learning resources and cited research. Hosting: Replit (Reserved VM) and Replit PostgreSQL 16; GitHub Actions runs a type-check and 543 tests. Flow: CV or quiz → local OCR and privacy scrubber → LLM and RunPod skill matching → PostgreSQL → fixed scoring, routes, Slow/Steady/Fast scenarios and SDG mapping → browser. Trusted market sources → quotation check → three-agent vote → person decides split votes → new knowledge version → keep-or-switch for affected users. (System block diagram: Appendix A.)

Sponsor resources used: Yes. RunPod for training and runtime embeddings; Replit for hosting the app, its PostgreSQL database and Secrets; KrackedDevs for the LLM proxy. Sign-in: Replit-managed Clerk, with guest access. Live app: https://halatuju.replit.app.

### 3.2 Design-Thinking & Deployment Strategy

*Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.*

***[Maximum 200 words]***

We framed the task around a sample mid-career accounts-payable worker with six hours a week and a six-month redeployment deadline. In the final round we reordered the product around her first question, "which of my skills will AI take?", before "where can I go?". Early HTML prototypes led to a visible map and route choices rather than an uncheckable chat answer. We tested CV extraction and privacy against held-out synthetic cases, then improved the scrubber after the blind first result. A fairness test checks that changing name, gender, age or religion changes nothing. The repo contains 543 automated tests, a type-check and documented evaluation results. Human approval covers split votes on route-changing evidence, and labelled fallbacks cover AI outages. The next step is a consent-based pilot with career centres and verified employer openings; parts of the catalogue remain labelled sample data. The app is live on Replit at https://halatuju.replit.app.

## 4. Preliminary Environmental & SDG Impact Framework

*To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.*

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess* ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

| **Metric Category** | **Your Response** |
|---|---|
| **Target SDG Sub-Target(s)**<br>*Identify the exact UN SDG sub-target your project addresses*<br>*e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."* | SDG 8.5: full and productive employment and decent work; SDG 4.4: skills for employment. The routes also support SDG 8.2 through technological upgrading. |
| **Current Problem Baseline**<br>*What is the starting condition of the problem, expressed as a number?*<br>*e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."* | Context baseline: 1,360,100 Malaysian clerical wage recipients in 2025 (DOSM/OpenDOSM, wages_occupation). Product baseline on a held-out synthetic set: 97.2% correct current-job identification (35/36); 99.4% planted personal details removed after fixes. These are test results, not measured labour impact. |
| **Core Measurable KPI**<br>*The specific unit you will track*<br>*e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.* | Track (1) % correct current-job identifications on new CVs; (2) % planted personal details removed before AI calls; (3) % users who finish a costed route within 60 seconds; (4) % pilot participants with an approved redeployment plan by their deadline. |
| **Phase 2 Impact Target**<br>*Your realistic, percentage-based target by the final prototype presentation*<br>*e.g., "20% improvement in computational resource use."* | Phase 2 targets: 95% current-job identification and 99% PII removal on a fresh blind set; 80% of pilot users finish a route in 60 seconds; 70% of consenting pilot participants have an advisor-approved plan by their deadline. Pilot targets are proposed, not achieved. |

*This baseline is kept exactly as submitted on 25 October. Final-round results against it are in Section 4 of SDC2026_Dsaise_Final.pdf.*

## 5. Eligibility & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

## 6. IP & Originality Declaration

### 6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

### 6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

| Name | Type (code/dataset/model) | License |
|---|---|---|
| Node.js, Express, TypeScript, Tesseract, PostgreSQL libraries, 3D graph and GSAP | Code | MIT / Apache-2.0 / BSD / GreenSock terms; full inventory in technical declaration |
| Google Gemini and BAAI/bge-m3 | Models | Gemini API terms; MIT |
| ESCO v1.2.1 and ESCO–O*NET crosswalk; O*NET | Datasets | European Commission reuse policy; O*NET CC BY 4.0 |
| ILO WP140, Anthropic Economic Index, GPTs are GPTs | Research data | CC BY 4.0 / CC BY / MIT as applicable |
| DOSM/OpenDOSM, MyCOL and public course listings | Datasets | CC BY 4.0 / facts with attribution / linked provider facts |

| | | |
|---|---|---|
| Clerk SDK | Code | MIT |
| GPT-5.5, gpt-6-sol, DeepSeek-V4-Pro via the KrackedDevs proxy | Models (API) | Provider and sponsor terms |
| O*NET 30.2 Detailed Work Activities | Dataset | CC BY 4.0 |
| Learning resources (YouTube, Microsoft Learn, IIA, GHG Protocol, IFRS Foundation and others), links only | Dataset (links) | Provider terms; linked only |
| UN SDG icons | Graphics | UN SDG icon guidelines |

**6.3 Employment-Related IP Conflicts**

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

## 7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Date: 31 October 2026

## Appendix A: System block diagram

